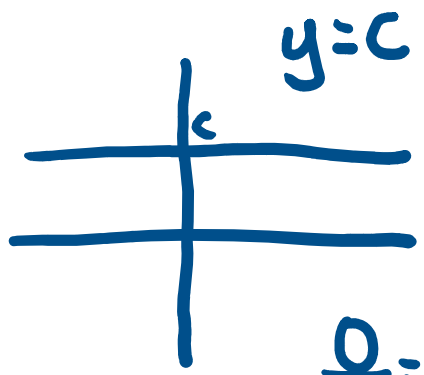
When we have a constant equation, f(x)=c and f(x+Δx)=c

We put that in for our generic functions.

$$\frac{f(x+\Delta x)-f(x)}{\Delta x}$$



$f(x)=c$

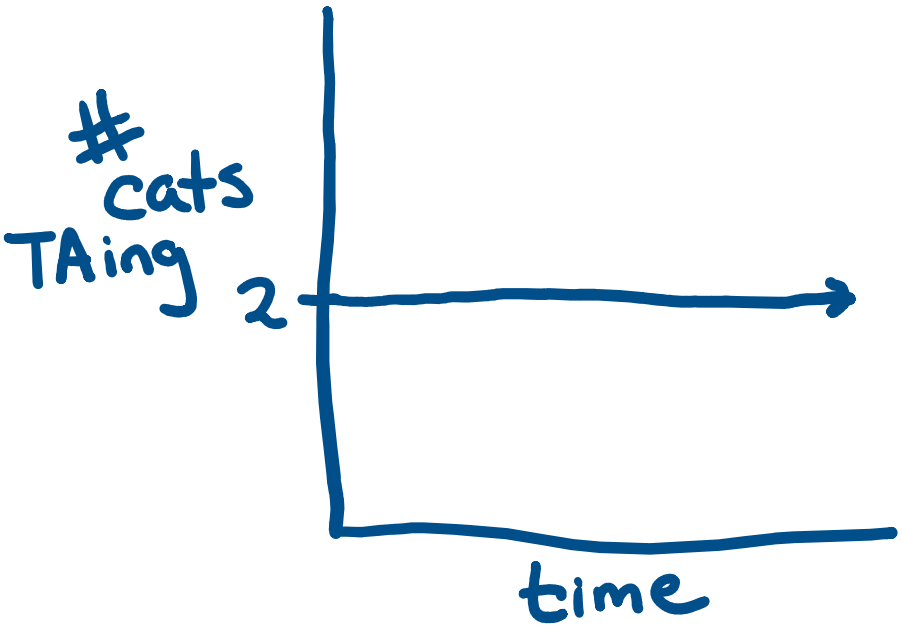
$f(x+\Delta x)=c$

$$\frac{0}{n}=0$$

$$\frac{0}{\Delta x}=0$$

$$\frac{c-c}{\Delta x}=\frac{0}{\Delta x}$$

Here f(x+ Δx)-f(x) is zero, the slope is zero and the derivative is zero.



$$\frac{2-2}{\Delta t} = 0$$

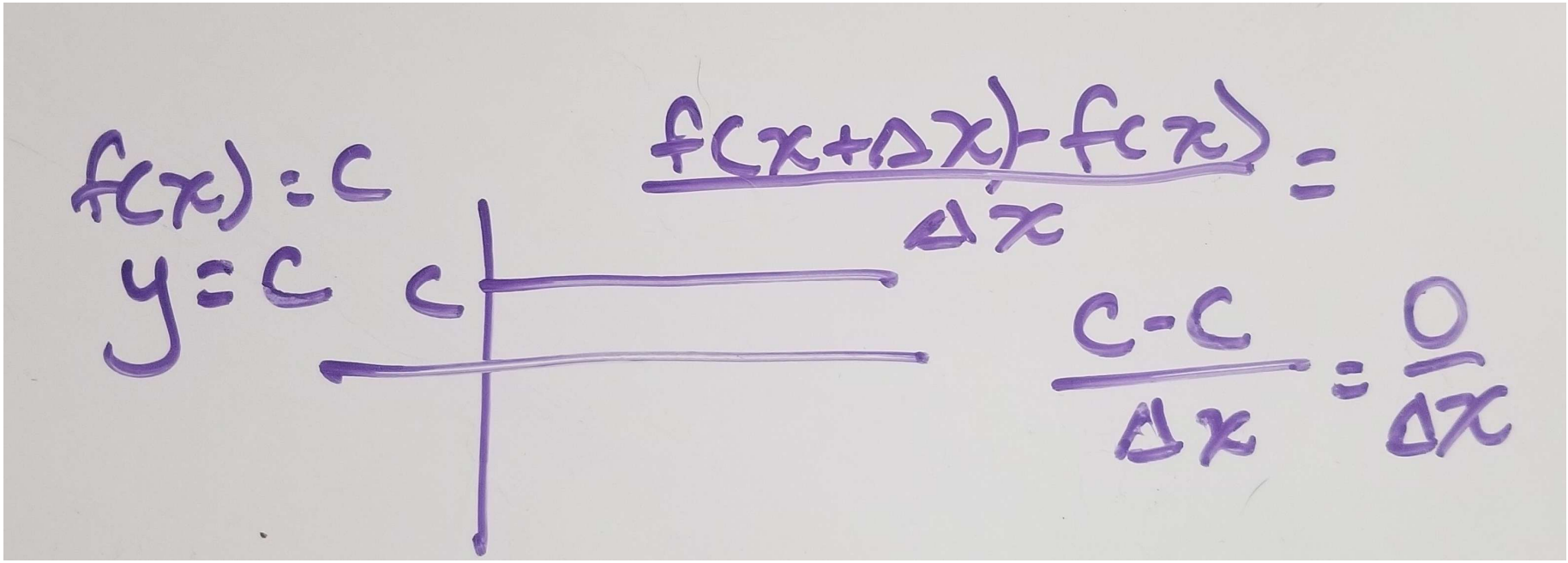
$$f(x) = 0x + 2$$

$$f(x) = 2$$

$$f(x+\Delta x) = 2$$

$$\frac{2-2}{\Delta x} = 0$$

The derivative of a function is the rate of change of the output of the function at any given input.



The slope or rate of any constant function or term is 0 so the derivative of any constant function or term is 0.

$y = 2$

$y' =$

$y = 5$

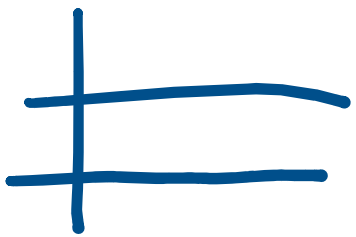
$\frac{dy}{dx} =$

$y = -3$

$y' =$

$f(x) = 27.893$

$f'(x) =$



$y = -7$

$\frac{dy}{dx} =$

$f(x) = \pi$

$f'(x) =$

$f(x) = 30.5$

$f'(x) =$

When we have a linear equation, $f(x)=mx+b$ and $f(x+\Delta x)=m(x+\Delta x)+b$

We put that in for our generic functions.

$$\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

$$\frac{(m(x+\Delta x)+b)-(mx+b)}{\Delta x}$$

Distribute out the ms and the negative sign.

$$\frac{(m(x+\Delta x)+b)-(mx+b)}{\Delta x}$$

$$\frac{mx+m\Delta x+b-mx-b}{\Delta x}$$

Cancel out the like terms. Notice a pattern here? The new function minus the old function leaves the slope times delta x.

$$\frac{\cancel{mx} + m\Delta x \cancel{+b} - \cancel{mx} \cancel{-b}}{\Delta x}$$

$$\frac{m\Delta x}{\Delta x}$$

$$\frac{\Delta x}{\Delta x} = 1$$

Notice that b (which is the constant term, the y-intercept, or the starting point of the function, y when x is zero) cancels out and goes away. We said that we don't care about the y-intercept, or the value of the original output, just the slope of the line, which is the rate of change at that x.

$$\frac{\cancel{mx} + m\Delta x \cancel{+b} \cancel{-mx} \cancel{-b}}{\Delta x}$$

$$\frac{m\Delta x}{\Delta x} = m$$

Cancel out the delta x because any number over itself is one, which is the identity for multiplication so we can make it invisible.

$$\frac{m\cancel{\Delta x}}{\cancel{\Delta x}} = m$$

Practice taking the derivative of a constant or linear term.

$y = 10x$

$y' =$

$y = 5x$

$y' =$

$y = 3$

$y' =$

$y = 2x + 5$

$y' =$

$f(x) = -3x$

$f'(x) =$

$f(x) = 8x$

$f'(x) =$

$f(x) = -5$

$f'(x) =$

$f(x) = \pi$

$f'(x) =$

$y = x$

$\frac{dy}{dx} =$

$y = -45$

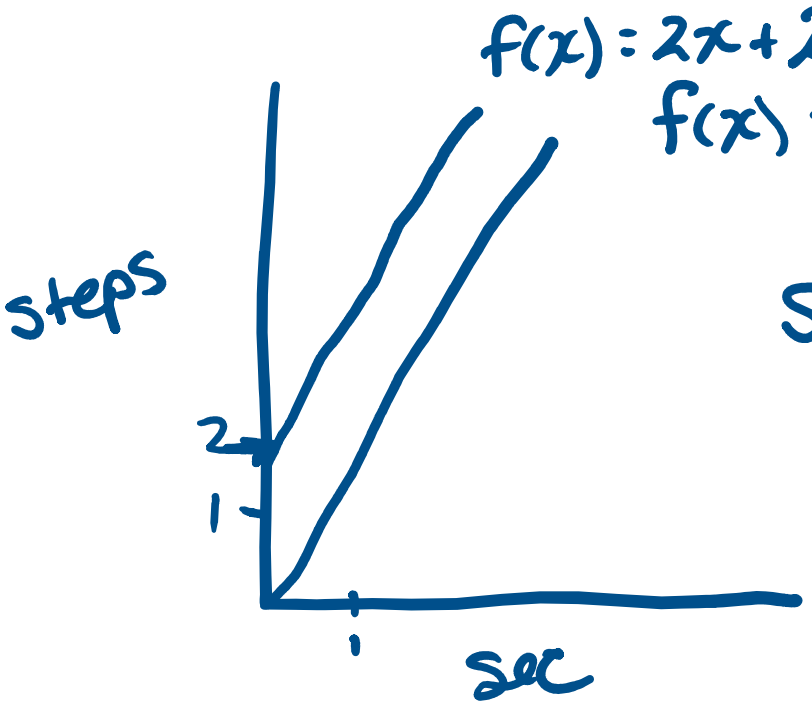
$\frac{dy}{dx} =$

$p = 60t$

$\frac{dp}{dt} =$

$\frac{dp}{dx} = ?$

When we take the derivative, we lose information about the original output. We also change our units.



$$\text{Steps} = 2\,\frac{\text{steps}}{\text{sec}} \cdot \text{sec} + \text{starting steps}$$

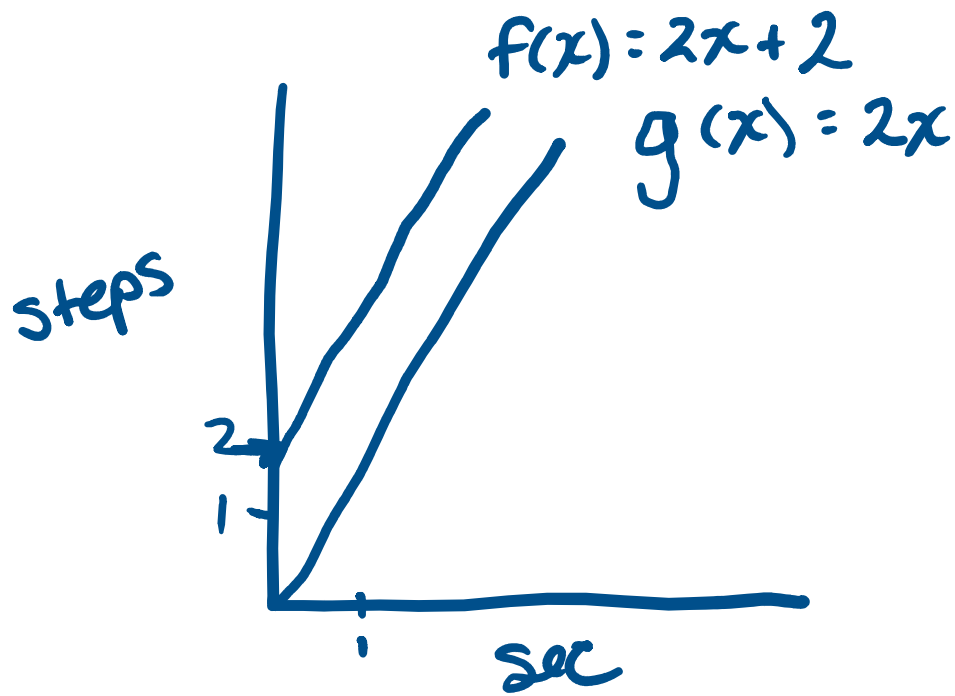
$$f(x) = 2x + f(0)$$

$$f'(x) = 2\,\frac{\text{st}}{\text{sec}} + 0$$

$$y = 2x + y_0$$

$$y' = 2\,\frac{\text{st}}{\text{sec}} + 0$$

The derivative as a function that takes an input function and outputs a new function that is many-to-one. Many different functions can give the same derivative. This will be important to remember when we reverse the process, which is called taking the integral.



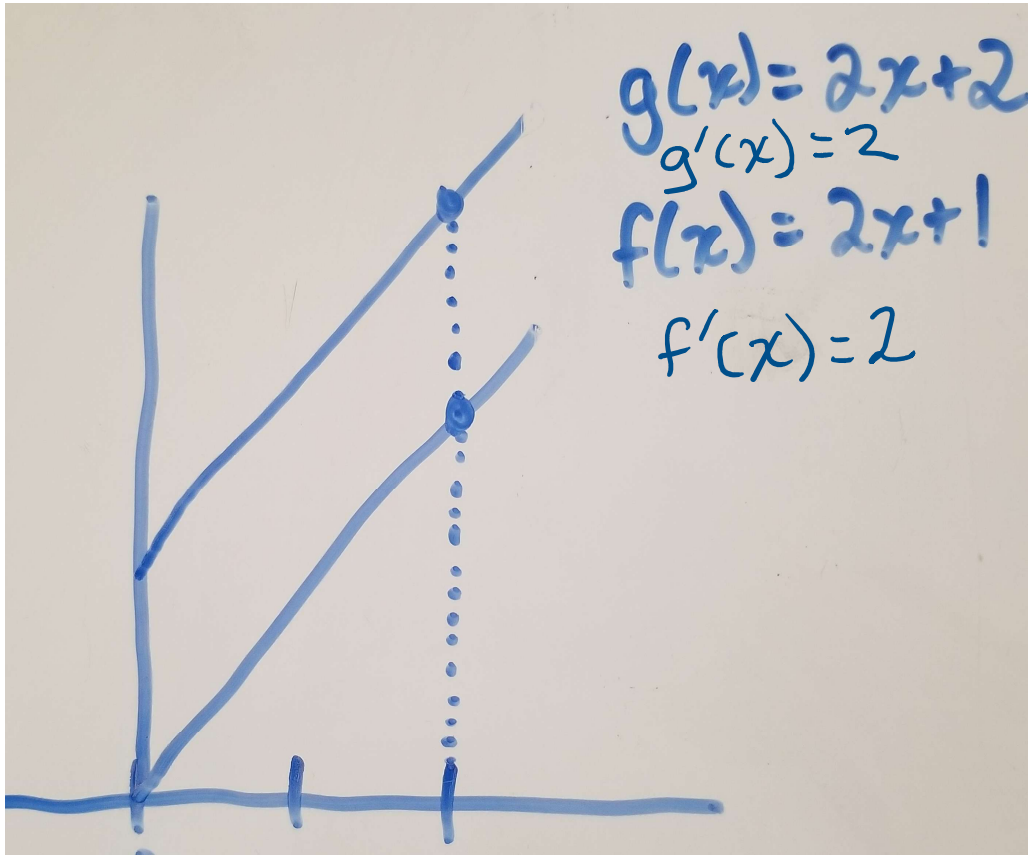
When we take the derivative, we lose information about the original output. We also change our units. The derivative is the units of the slope or the rate.

Miles= r miles/hr * hr +starting miles

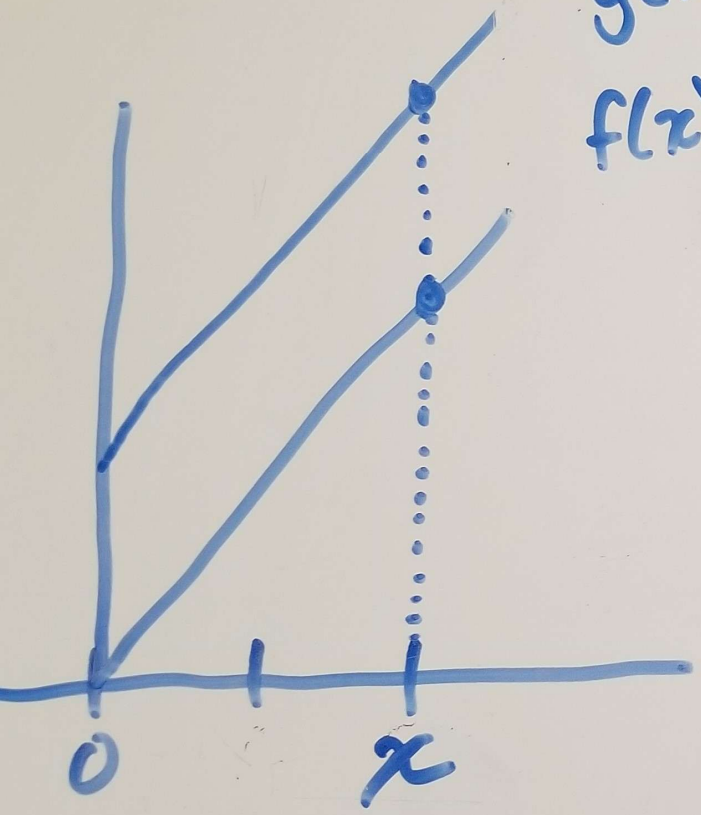
The derivative is r miles/hr

$$\frac{\Delta y}{\Delta x} \qquad \frac{\text{change output}}{\text{change input}}$$

Here we have two original linear functions with the same slope but different y-intercepts.



Same slope, parallel lines, the rate of change is the same at each point on both lines and is 2 in whatever units the rate would be.



$g(x) = 2x + 2$ $\quad g' = \dfrac{2(x+\Delta x)+2-(2x+2)}{\Delta x}$ $\quad g'(x) = 2$

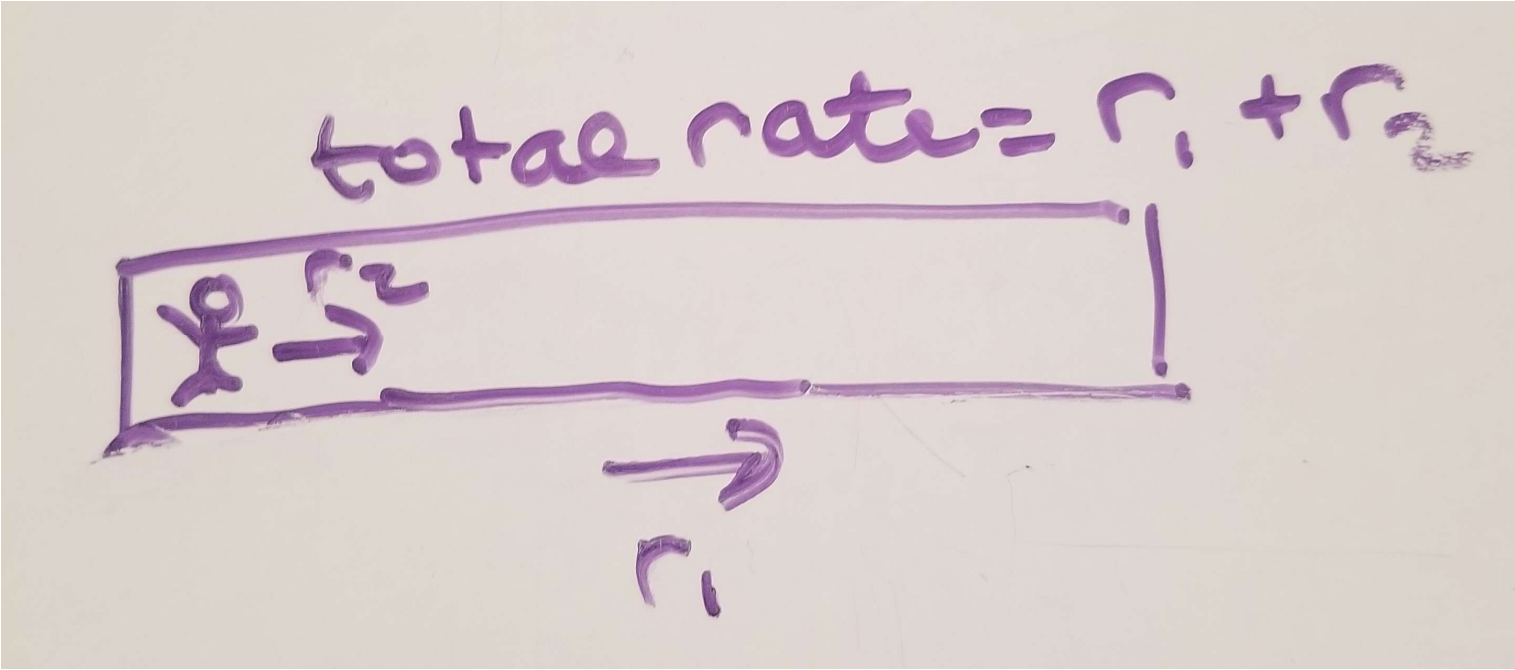
$f(x) = 2x + 1$ $\quad f' = \dfrac{2(x+\Delta x)+1-(2x+1)}{\Delta x}$ $\quad f'(x) = 2$

$g(x) = 2x + 2$ $\quad f(x) = 2x + 1$

$g'(x) = 2$ $\quad f'(x) = 2$

Rates add. ("As long as you are not near the speed of light." –Alan)

If you are on a train, in a car of the train and walking from the back to the front of the train car going 1 mph and the train is going 30 mph, you are going 31 mph in relation to the ground.



The derivative distributes over addition.

y=2x+3

y'=2+0 = 2

$$\frac{d(2x+3)}{dx} = \frac{d(2x)}{dx} + \frac{d(3)}{dx}$$

f(x)=g(x)+h(x)

f'(x)=g'(x)+h'(x)

g(x)=2x, h(x)=3, f(x)=2x+3

g'(x)=2, h'(x)=0 f'(x)=2+0 = 2

$y = 2x + 3x = 5x$

$y' = 2 + 3 = 5$

If you are adding functions to get a total function, you can add the derivatives to get the total derivative.

total = part1 + part2 + ...

$f(x) = ax + b$

$f'(x) = a + 0$

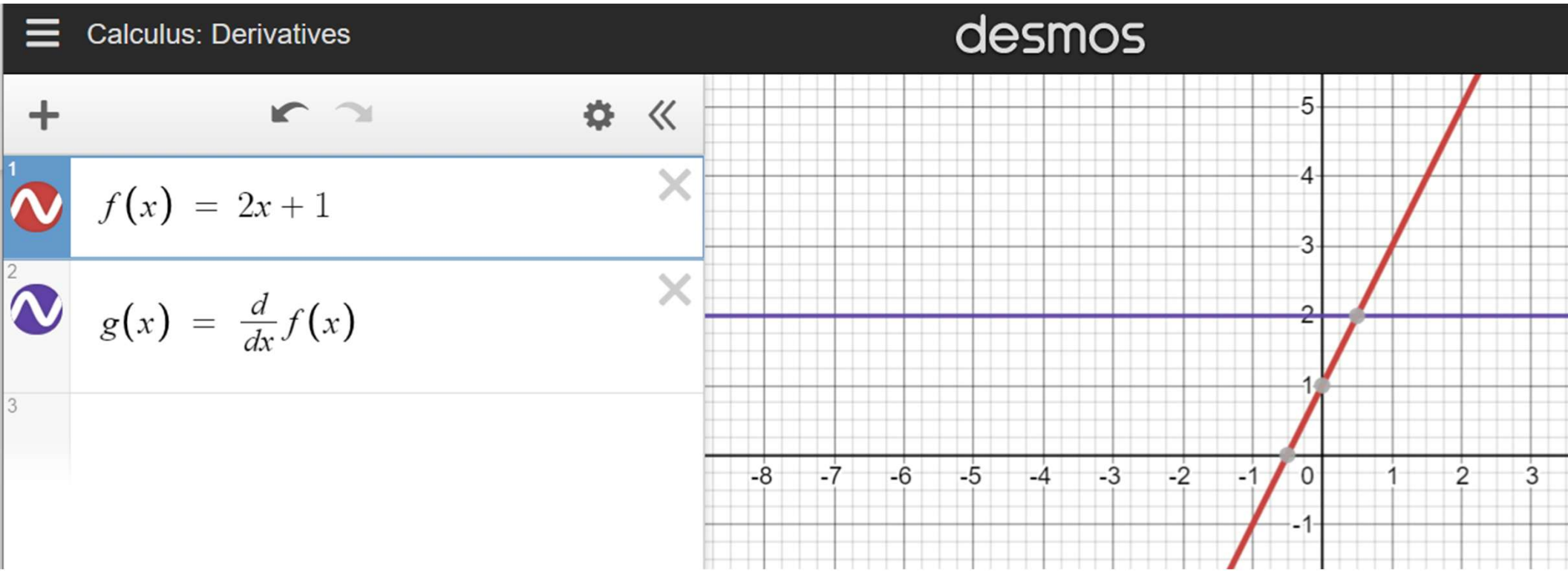
$y = ax + bx + cx + d$

$y' = a + b + c + 0$

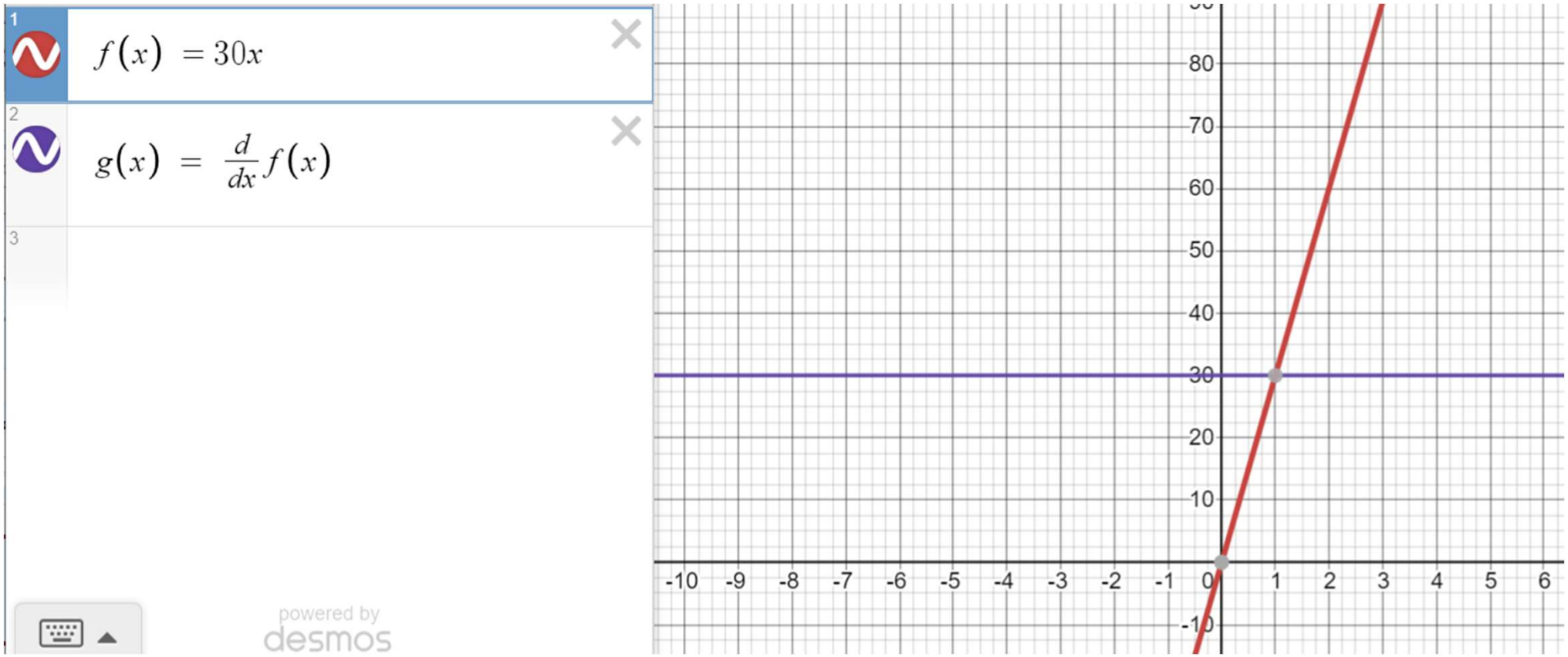
$y = 2x + 3x + 5x + 23$

$y' =$

Derivatives on Desmos



With the going 30 mph example



The slope of a constant function or term is zero for all input, and the derivative is zero.

y=5
y’=0

y=-3
y’=
y=4.5
y’=
y=3456789.43562

y=π

The derivative of a linear function or term is the slope, rate, or the coefficient of the linear term.

$y=3x+5$

$y'=$

$y=45.3x+2$

$y'=$

$y=4x$

$y'=$

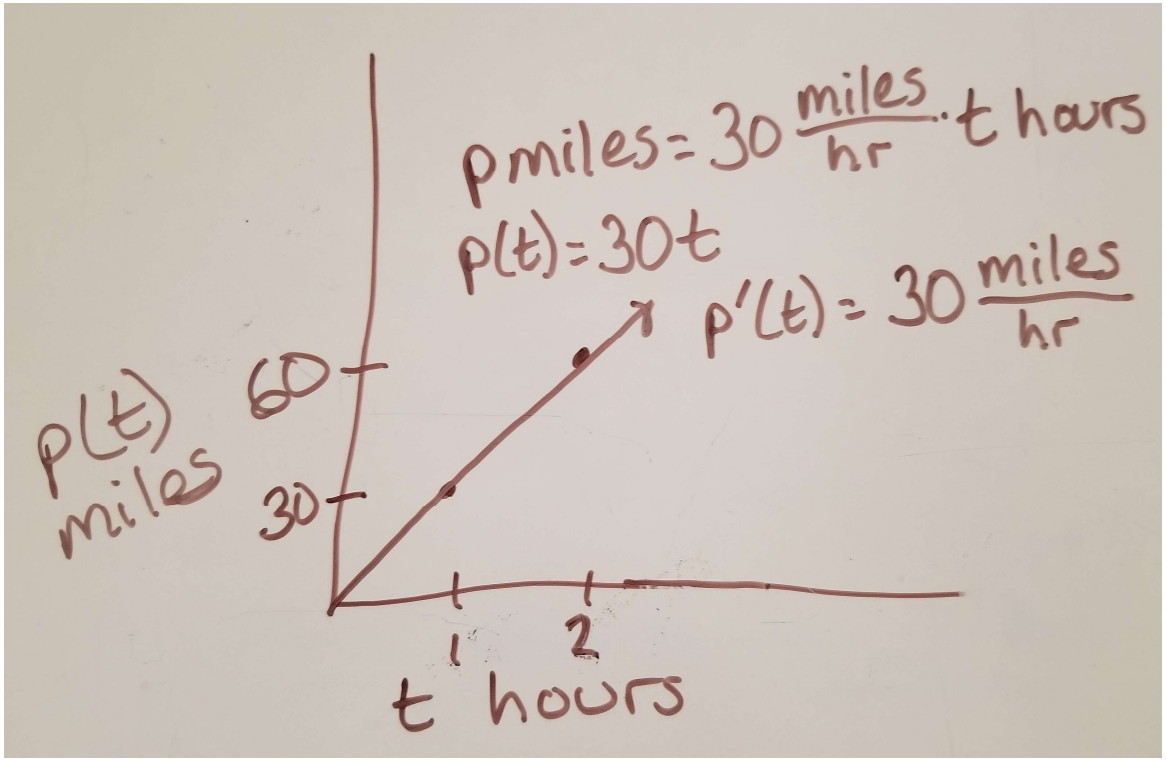
$y= -8x$

$y'=$

$y= -3.2x+4$

$y'=$

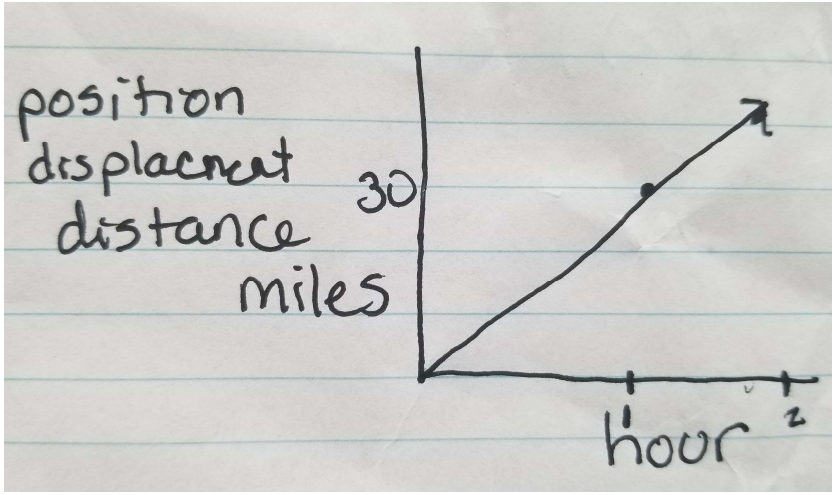
The derivative is a rate and so it in the units of the rate.

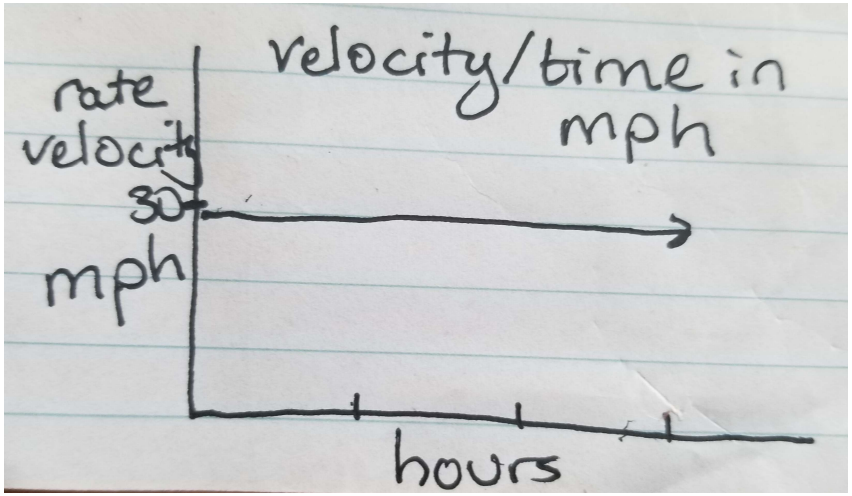


Here the units are input of hours and output in miles

Here the units are input of hours and output of miles per hour.

The units of the derivative are the units of the slope of the original function.





The derivative is the rate of change of the output of the function at any given point or input.

The derivative of a constant is zero.

$y=c$

$y'=0$

If you are at the same position, then you aren't going any distance in time and your speed is zero.

For polynomial, the derivative of the linear term is the coefficient, rate, multiplier or slope of the linear term.

$y=ax$

$y'=a$

If you are going a constant speed, then the rate at any time is the speed that you are going.